

# Joonsuk Bae

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*Lead analyzer of the first ALICE Run 3 charged-particle jet measurement. Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.*

## Education

**Ph.D. Candidate, Physics**, Sungkyunkwan University 2022.03 – expected 2026.08  
Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp at  $\sqrt{s} = 13.6$  TeV with ALICE.*  
**B.S., Physics**, Sungkyunkwan University 2021.02

## Appointments and Collaboration Membership

ALICE Collaboration, CERN 2022.10 – present  
ePIC Collaboration, Brookhaven National Laboratory 2024.02 – present  
LAMPS Collaboration, RAON 2022.03 – 2024.04

## Research Highlights

- **Lead analyzer**, first ALICE Run 3 charged-particle jet cross-section measurement at  $\sqrt{s} = 13.6$  TeV (in preparation, target 2026): end-to-end pipeline from O<sup>2</sup>Physics task development through QA, unfolding, and full systematic-uncertainty programme; PWG-JE-approved results establish the Run 3 inclusive charged-jet baseline.
- **PWG-JE Run 3 tracking-systematics task force** (convened 2026.02 by H. Caines and L. Havener, Yale; biweekly): pp  $\sqrt{s} = 13.6$  TeV lead analyzer. Derived the total charged-particle tracking-efficiency systematic ( $\sim 1.5\%$  across the dominant  $p_T$  range;  $p_T$ -binned), including extension to the 2022 and 2024 periods. ITS-TPC matching code adopted by J. Lee for PbPb tracking. Findings feed the Hard Probes 2026 systematic-uncertainty proposal.
- **ALICE O<sup>2</sup>Physics framework** (AliceO2Group/O2Physics): 35 pull requests / 19 merged including a central patch in Common/ (PR #15174, TrackTuner) and sole author of `jetCrossSectionEfficiency.cxx` (PR #14057) used by all Run 3 charged-jet analyses for cross-section normalization.
- **ePIC Barrel Imaging Calorimeter**: lead analyzer for the on-site prompt and follow-up offline analysis of the CERN PS T10 (2025.07) Pb/scintillating-fiber test beam; co-author on the first beam-test paper (submitted, NIMA-D-26-00482) with module-construction support and PMT/module performance characterization.
- **Mentoring of MSc analyzers** (W. Ham + 1) on the ALICE Run 3 O–O charged-jet measurement: jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author on the corresponding Analysis Note with service-level normalization contribution.

## Selected Publications

Full list at [inspirehep.net/authors/2117232](https://inspirehep.net/authors/2117232); ALICE Collaboration co-author since 2022.10 (**109 collaboration papers**).

- **FLAGSHIP** ALICE Collaboration, *Inclusive and semi-inclusive charged-particle jet cross sections in pp collisions at  $\sqrt{s} = 13.6$  TeV* (in preparation, target 2026). [*lead analyzer; first ALICE Run 3 charged-jet publication*]
- Korea-BIC Collaboration, *Beam-test characterization of a Pb/SciFi BIC prototype — CERN PS T10 (2025.07)* (in preparation, 2026). [*lead analyzer; prompt and follow-up offline analysis*]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae et al.**, “Beam test of a Pb/SciFi prototype for the BIC at the EIC,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; arXiv preprint). [*co-author; module support and PMT/module performance characterization*]
- Y. Hong, J. Bok, G. An, **J. Bae et al.**, “Test-beam performance of the AstroPix silicon sensor for imaging calorimetry,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). [*co-author; ePIC BIC silicon-layer test beam*]
- J. K. Adkins et al. (ECCE Coll.), “Design of the ECCE detector for the EIC,” *Nucl. Instrum. Meth. A* **1073** (2025) 170240. [*co-author; ECCE consortium*]
- Y. J. Kim et al. (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682. [*co-author; Time-of-Flight maintenance and on-site checks*]

## Selected Talks (since 2022)

**Summary:** 3 invited workshops · 5 invited collaboration-meeting talks · 6 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025) · 7 domestic (KPS) · PWG-JE internal: 50+.

- **EPS-HEP 2025**, Marseille [on behalf of the ALICE Collaboration; PWG-JE merge talk] — 2025.07
- **HIM 2025**, APCTP [invited workshop] — *From precision to structure: jet measurements in ALICE Run 3* — 2025.05
- **Quark Matter 2025**, Frankfurt [international, poster] — 2025.04
- **ALICE Physics Week 2024** [invited collaboration meeting] — *Charged-particle jet production in Run 3* — 2024.12
- **Hard Probes 2024**, Nagasaki [international, poster] — 2024.09

## Software and Computing

ALICE O<sup>2</sup>Physics (AliceO2Group/O2Physics, 2023 – present). 35 PRs / 19 merged, including:

- **Central framework patch in Common/** (PR #15174, merged) — TrackTuner  $Q/p_T$  smearing fix.
- **Sole author of** `jetCrossSectionEfficiency.cxx` (PR #14057, merged) — PWG-JE task used by all Run 3 charged-jet analyses for normalization.
- Code review on 5 community PRs across PWG-JE and Common.

## Detector and Test-beam Experience

- **ePIC Barrel Imaging Calorimeter** — on-site at **CERN PS T10** (2025.07; analysis lead) and **KEK AF-AR** (2025.03; DAQ shifter, prompt analysis); Pb/scintillating-fiber module-construction support and PMT/SiPM-level QA across all prototype modules; JANA2 / EIC-recon reconstruction studies for sampling-fraction and energy response.
- **ALICE ITS2 alignment monitoring** (CTF level) — service work providing period-by-period tracking-performance diagnostics from compressed time-frame data, below the standard AO2D analyzer interface.
- **LAMPS Forward Tracker** (2022 – 2024) — MPPC/scintillator-plane forward-tracking concept design and Geant4 feasibility simulations.

## Service and Mentoring

- **Student mentoring** (non-supervisory) — W. Ham and one further MSc co-analyzer on the ALICE Run 3 O-O charged-jet measurement: jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author on the corresponding Analysis Note.
- **KoALICE O<sup>2</sup> Tutorial** (2024.01.22–23, SKKU; co-organizer with H. Lee and listed Indico contact) — two-day Run 3 analysis-framework tutorial for the Korean ALICE community; instructor on *Running O<sup>2</sup>Physics with Hyperloop* (solo) and on *PWG-JE practical tasks*.
- **ePIC Korea-BIC analysis coordination** — primary liaison between the SKKU group and the Korea-BIC consortium for beam-test data products.

## Technical Skills

**Programming.** C++, Python, ROOT, Geant4, JANA2 / EIC-recon, PYTHIA, FastJet, Bash, Git, L<sup>A</sup>T<sub>E</sub>X.

**Analysis.** jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.

**Detector hardware.** PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

**Languages.** English (fluent), Korean (native).

## References

Available on request.